PROBLEM SET 1

NUMBER SYSTEM

1. Convert the following binary numbers to decimal.
(a) 1 0 11 (b) 1 1 0 1 1 0 1 (c) 1 1 0 1 . 1 1 (d) 1 1 0 1 1 1 0 . 0 1 1

2. Convert the following decimal numbers to binary

(a) 37 (b) 28 (c) 197.56 (d) 205. 05

3. Add the following binary numbers.
(a) 11011 + 1101 (b) 1011 + 1101 + 1001 + 1111
(c) 10111. 101 + 110111. 01 (d) 1010.11 + 1101.10 + 1001.11 + 1111.11

4. Subtract the following binary numbers.
(a) 1011 – 101 (b) 10110 – 1011 (c) 1100.10 – 111.01 (d) 10001.01 – 1111.11

5. Convert the following hexadecimal numbers to octal.

(a) 2AB (B) 42FD (c) 4F7.A8 (d) BC70.0E

6. Convert the following octal numbers to decimal

(a) 463 (b) 2056 (c) 2057.64 (d) 6534.04

7. Convert the following decimal numbers to octal.

(a) 287 (b) 3956 (c) 420.6 (d) 8476.47

8. Add the following octal numbers.

(a) 173 + 265 (b) 1247 + 2053 (c) 25.76 + 16.57 (d) 273.56 + 425.07

9. Subtract the following octal numbers.

(a) 64 – 37 (b) 462 – 175 (c) 175.6 – 47.7 (d) 3006.05 - 2657.16

10. Convert the following hexadecimal numbers to binary.

(a) C20 (b) F297 (c) AF9. B0D (d) E79A.6A4

11. Convert the following binary numbers to hexadecimal.

(a) 10110 (b) 1011011011 (c) 110110111.01111 (d) 1101101101101.101101

12. Convert the following hexadecimal numbers to decimal.

(a) AB6 (b) 2EB7 (c) A08F.EA (d) 8E47.AB

13. Add the following hexadecimal numbers.

(a) AC6 + B59 (b) A0FC + B75F (c) E0F3.5D + 49E6.F7 (d) ABC7.54 + 26F3.AB + DAC9.6F

14. Subtract the following hexadecimal numbers.

(a) BC5 – A2B (b) F27 – B9E (c) CDF7.52- AB5.8 (d) 67F2.6E – 4A0E.A9